

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Heat the ends of the rods [where they come together equally with a Bunsen flame] (1) Pin drops off (1) Either: <u>And</u> {the best conductor / copper} is first (1) Or: Vaseline melts first on copper (1)	3			3		3
		(ii)		Particles vibrate <u>faster</u> and collide with their neighbours (1) [Free] <u>electrons</u> move through the structure [passing on their energy] (1)	2			2		
	(b)	(i)		Hot water rises (1) Because it {is less dense / expands} (1) Accept converse argument. Treat as neutral any reference to purple crystals	2			2		2
		(ii)		Hot air {above the radiator / at the top of the room} rises (1) So the air below is not heated by <u>convection</u> (1)		2		2		
				Question 4 total	7	2	0	9	0	5